Skill Forecasting and Semantic Career Pathway Optimization Ecosystem for Undergraduates

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Project Proposal Report

S. R. P. A. M. D. N. Alphonso – IT22153890

B.Sc. (Hons) Degree in Information Technology Specialized in Information Technology

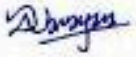
Department of Information Technology

Sri Lanka Institute of Information Technology Sri Lanka

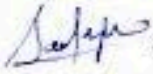
March 2026

DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

Name	Student ID	Signature
S. R. P. A. M. D. N. Alphonso	IT22153890	

The above candidate is conducting research for their undergraduate Dissertation under my supervision.

Name	Date	Signature
Ms. Sadeepa Dushanthi Kuruppu	13/03/2026	

ABSTRACT

The transition from university to professional employment in Sri Lanka's IT industry is a challenging phenomenon, and existing systems of career guidance lack dynamic models of career development in predicting the future of IT undergraduates in Sri Lanka. Career guidance systems currently in use only allow for static models of job recommendation and do not cater to dynamic models of career development. The proposed research will address this issue of predicting career pathways for Sri Lankan IT undergraduates by developing a sequence modeling system for predicting probable career paths of IT undergraduates in Sri Lanka. The proposed component of this research will develop a component for predicting probable career paths for IT undergraduates in Sri Lanka by using a sequence modeling system based on transformer models and developing a component for visualizing probable career paths for IT undergraduates in Sri Lanka. The expected outcome of this research is that a system for predicting probable career paths for IT undergraduates in Sri Lanka will be developed, and its accuracy in predicting probable positions will be more than 80%. Additionally, an interactive graph will be developed for visualizing probable career paths for IT undergraduates in Sri Lanka.

Keywords: Career trajectory prediction, graph databases, sequence modeling, IT workforce, probabilistic pathways

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
AI	Artificial Intelligence
API	Application Programming Interface
ARIMA	AutoRegressive Integrated Moving Average
BERT	Bidirectional Encoder Representations from Transformers
RP	Research Project
CTP	Career Trajectory Prediction
CoEAI	Centre of Excellence for Artificial Intelligence
CV	Curriculum Vitae
DNN	Deep Neural Network
ESCO	European Skills, Competences, Qualifications and Occupations
GDPR	General Data Protection Regulation
GNN	Graph Neural Network
ICT	Information and Communications Technology
IEEE	Institute of Electrical and Electronics Engineers
IT	Information Technology
JSON	JavaScript Object Notation

KDD	Knowledge Discovery in Databases
KG	Knowledge Graph
LMSI	Labour Market and Skills Intelligence
MOOC	Massive Open Online Course
Neo4j	Network Exploration and Optimization 4 Java
NLP	Natural Language Processing
OCR	Optical Character Recognition
REST	Representational State Transfer
RNN	Recurrent Neural Network
SDG	Sustainable Development Goals
SLIIT	Sri Lanka Institute of Information Technology
SVM	Support Vector Machine
TKG	Temporal Knowledge Graph
UI	User Interface
WBS	Work Breakdown Structure

1. INTRODUCTION

1.1 Background and Context

The Real-World Problem:

The critical challenge faced by IT graduates in Sri Lanka today is the lack of visibility of the potential career paths that match their skills and interests. Unlike other professions, such as medicine and law, where there are clear career path advancements, the IT industry offers highly divergent career paths. A software engineer can potentially move into technical architecture, engineering management, solutions consulting, or any of the various domains in the industry, such as data science or cybersecurity. The reality today is that students make these critical decisions based on very little information - rumors and hearsay from seniors in the industry, general information from the Internet, or job titles that do not necessarily reflect the actual job functions.

Who is Affected and Why It Matters:

There are three main groups of people who are impacted by the problem. Undergraduate IT Students face decisions about specializations, electives, and internship applications without a clear understanding of the long-term implications of these decisions. University Career Guidance Units face difficulties in offering informed advice to their students without the ability to model relationships between skills, roles, and career outcomes. Employers face difficulties in workforce planning and retaining staff without the ability to model career paths.

Statistical Evidence:

According to recent labor market analysis, it is estimated that 20-25% of IT graduates in Sri Lanka suffer from underemployment after graduation. The IT industry contributes 5% to Sri Lanka's GDP, and skills shortages in this industry hinder its growth.

SDG Alignment:

This research supports:

- SDG 4 (Quality Education) – Enabling evidence-based educational decisions
- SDG 8 (Decent Work) – Reducing skill mismatches and promoting productive employment

Research Cluster Alignment:

This project aligns with the FoC Research Cluster Centre of Excellence for AI (CoEAI) and intersects with the industry verticals of Education, Technology, and Human Resource Management.

1.2 Literature Review

Career Trajectory Prediction Research

The first career prediction models were based on Markov models that used fixed probability patterns for career transitions. However, these models were not effective as they failed to incorporate the various dependencies associated with previous positions and skills.

Neural Sequence Models

Vafa et al. proposed an extension to the existing sequence model based on the transformer neural network architecture for job sequences. They proposed two-stage predictions for career transitions and understanding the impact of early career decisions on subsequent ones. They used 24 million career trajectories for American workers and achieved better results than the existing models.

Knowledge Graph Models

Li et al. proposed a graph-based data repository for career-based course recommendation. They used Neo4j for creating relationships between careers, courses, and skills. They proposed the CAPER system for career prediction based on temporal knowledge graphs for understanding relationships between users, positions, and companies over time. This model was found to provide 34.58% more accurate results than existing career prediction models.

Skill-Based Career Modeling

Gollapalli et al. proposed a model based on the idea that skills are more consistent than job titles for users across various organizations. They proposed a model based on unified skill graphs

1.3 Research Gap

There is no validated localized career path prediction system for IT undergraduates in Sri Lanka, where graph-based trajectory modeling is combined with probabilistic sequence prediction and provides recommendations.

Existing career path prediction systems answer "What jobs are available now?" but not "What career paths are possible from where I am, and what is the probability of these paths?" The research-based career path prediction systems cannot be used directly for Sri Lanka due to issues such as data distribution, scale, and context for IT-related jobs.

2. OBJECTIVE

2.1 Main Objective

To build a graph-based system for predicting the career pathway of Sri Lankan undergraduates by identifying the possible IT career pathways by developing a temporal knowledge graph construction system coupled with a probabilistic sequence prediction system to attain >80% accuracy for predicting the next position in the career pathway.

2.2 Specific Objectives

1. To construct a temporal knowledge graph of IT career trajectories by collecting and anonymizing career histories from Sri Lankan IT professionals, representing relationships between individuals, positions, and organizations with temporal attributes.
2. To design a graph database schema in Neo4j that efficiently stores career trajectory data and enables path analysis queries.
3. To develop a transformer-based sequence prediction model that learns transition probabilities from career sequences using two-stage prediction (transition detection followed by role prediction).
4. To implement probabilistic path generation algorithms that generate multiple probable future career trajectories with associated transition probabilities and estimated timeframes.
5. To create an interactive visualization interface that presents predicted pathways as an explorable graph, allowing users to examine different branches and view transition probabilities.

3. METHODOLOGY

3.1 Research Approach

This research adopts the Design Science Research Methodology (DSRM), which is appropriate for developing and evaluating innovative IT artifacts to solve identified problems. The methodology comprises six activities:

1. Problem identification and motivation
2. Definition of solution objectives
3. Design and development
4. Demonstration
5. Evaluation
6. Communication

3.2 Technical Methods and Algorithms

Phase 1: Data Collection

Career histories will be collected from:

- LinkedIn profiles of Sri Lankan IT professionals (anonymized, public data only)
- University alumni records (de-identified, aggregated)
- Voluntary survey responses from industry professionals

Phase 2: Graph Construction

According to the CAPER paper, a temporal knowledge graph will be constructed to represent career histories as follows:

- Entity types: Person, Position, Company
- Relationship types: held_position, transitioned_to, employed_by
- Temporal attributes: start date, end date (for positions)

The graph database will be constructed using Neo4j due to its native support for property graphs and efficient path traversal.

Phase 3: Development of Sequence Model

The prediction model will be based on a transformer sequence model, and it will include features such as:

- Positional encoding of the stages of career development

- Two-stage output: Transition probability and conditional role distribution
- Transfer learning: Using pre-trained models on global data and then fine-tuning on local data in Sri Lanka

Phase 4: Path Generation

For a specific position, various paths will be generated through beam search, and only those paths will be considered if their probability is greater than a specific threshold value. The transition probability can be determined as follows:

$P(path) = \prod P(transition \mid history)$ for all steps in the path

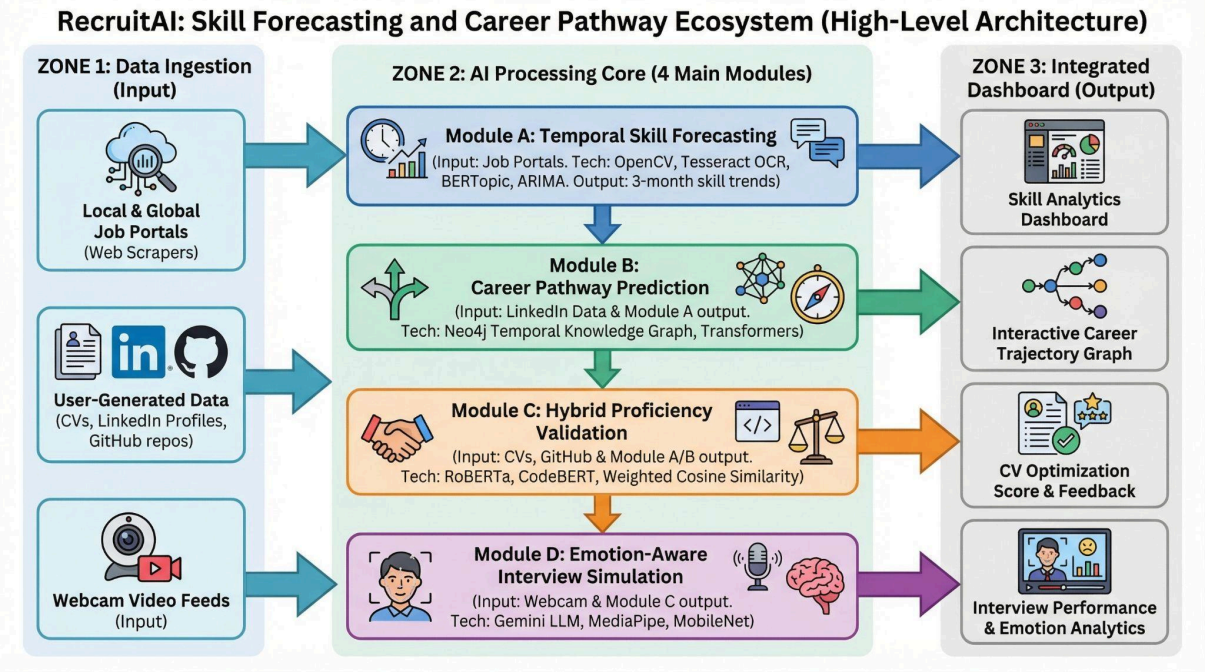
Phase 5: Visualization

A React-based interface will be created for visualizing the predicted paths in an interactive graph, and users will be able to click on specific nodes to get information on skills related to a specific role.

3.3 Tools and Platforms

Tool	Purpose
Neo4j	Graph database management
Hugging Face Transformers	Sequence model implementation
PyTorch	Deep learning framework
Python	Primary programming language
React.js	Frontend visualization
FastAPI	Backend API development

4. HIGH-LEVEL SYSTEM ARCHITECTURE



5. USER REQUIREMENTS

5.1 Stakeholder Identification

Stakeholder	Role	Needs
IT Undergraduate (Primary)	End user seeking career guidance	Understand possible career paths from current position; identify skill requirements for desired trajectories; compare pathway probabilities
Career Guidance Counselor	Advisor using system to support students	Generate pathway visualizations for advising sessions; access evidence-based career information; track student outcomes

University Administrator	Decision-maker for career services	Aggregate career outcome data; identify popular pathways; assess curriculum alignment with career patterns
HR Professional	Talent development specialist	Understand typical career progressions; benchmark internal career paths against industry patterns
Student (Exploring Options)	Prospective user	Explore career possibilities without commitment; understand required preparation for different paths

5.2 Functional Requirements

ID	Requirement	Priority	Stakeholder
FR-01	The system shall accept user input including current role, skills, education, and career preferences	High	Student, Counselor
FR-02	The system shall generate predicted career pathways up to 5 transitions from the current position	High	Student, Counselor
FR-03	The system shall display transition probabilities for each predicted move	High	Student, Counselor

FR-04	The system shall provide confidence intervals for predictions	Medium	Counselor, HR
FR-05	The system shall allow users to explore alternative paths by selecting different branches	High	Student
FR-06	The system shall display estimated timeframes for each transition	Medium	Student
FR-07	The system shall show skill requirements associated with each predicted role	High	Student
FR-08	The system shall highlight skill gaps between user's current skills and required skills for predicted roles	High	Student, CV Optimizer
FR-09	The system shall support filtering pathways by criteria (e.g., location, company size, industry)	Medium	Student
FR-10	The system shall provide explanations for predictions, showing similar career histories that support each path	Medium	Counselor, Student
FR-11	The system shall allow counselors to save and annotate pathways for specific students	Low	Counselor

FR-12	The system shall export pathway visualizations in common formats (PNG, PDF)	Low	Student, Counselor
FR-13	The system shall aggregate anonymized pathway data for institutional reporting	Low	Administrator
FR-14	The system shall update models monthly with new career history data	Medium	All

5.3 Non-functional Requirements

- Performance Requirements
- Usability Requirements
- Security Requirements
- Reliability Requirements
- Maintainability Requirements

6. COMMERCIALIZATION PLAN

6.1 Target Market and Customer Persona

Primary Market Segments

- Higher Education Institutions (Sri Lanka): 30+ universities and higher education institutions offering IT/CS degrees, each of which has a career guidance department looking for better tools for student guidance.
- IT Training and Bootcamp Providers: Training providers that want to show clear career paths for potential students.
- HR Tech Platforms: Job sites and HR tech companies looking for ways to integrate career paths into their solutions.
- Corporate HR Departments: IT companies (100+ in Sri Lanka) interested in workforce planning and internal mobility modeling.

Customer Personas

- Vidura - Career Guidance Counselor at a university in Sri Lanka: Needs evidence-based career information for her 200+ IT students every year. Currently only has her own experience and online information.
- Chamari - Final Year IT Student: Wants to know where her skills will take her and what skills she still needs for her dream job. Worried about making wrong career decisions.
- Kasun - HR Manager of an IT company in Sri Lanka: In charge of talent development and retention for the company. Needs to show her employees how they can move forward and what skills they lack.

6.2 Value Proposition

For Students:

- See your possible futures: Envisioning your possible career paths from your current position
- Make informed decisions: Data-driven decisions for your areas of specialization, electives, and career choices
- Identify your skill gaps: Knowing exactly what you need in your desired path
- Reduce uncertainties: Understanding the probability and timeframe for your possible futures

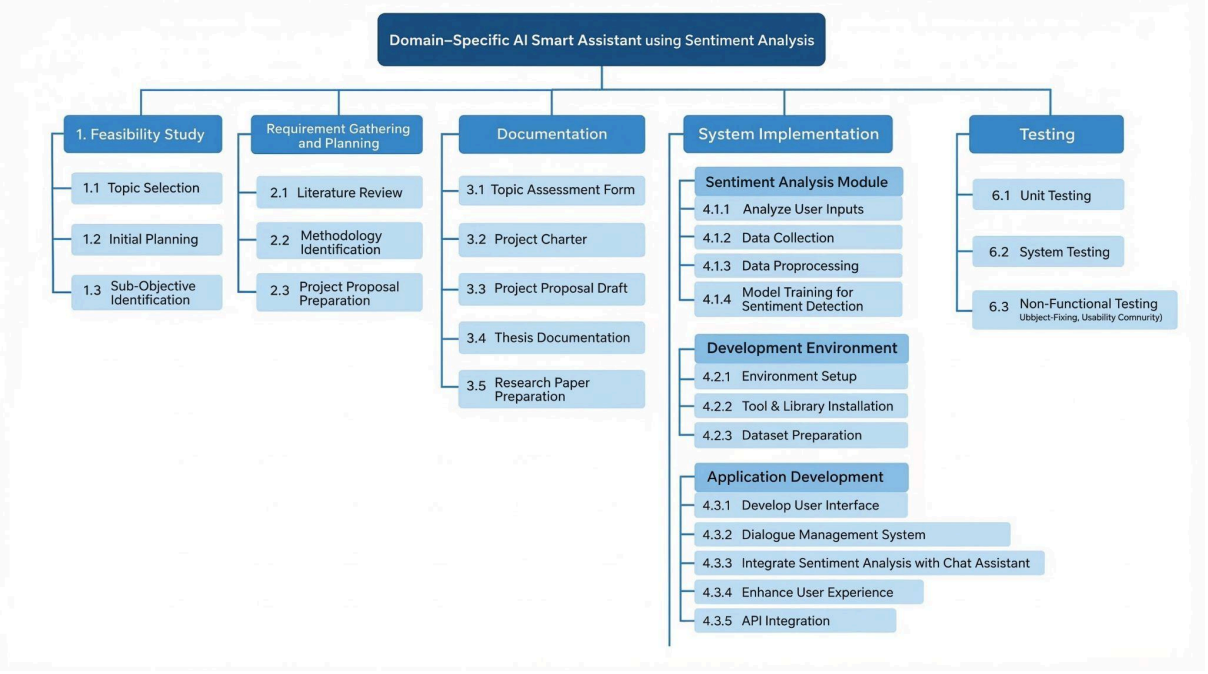
For Universities:

- Improve career services: Provide students with data-driven career guidance
- Demonstrate outcomes: Showcase career outcomes for prospective students through alumni career paths
- Curriculum alignment: Determine skills for successful career outcomes
- Competitive differentiation: Provide students with cutting-edge career tools not found anywhere else

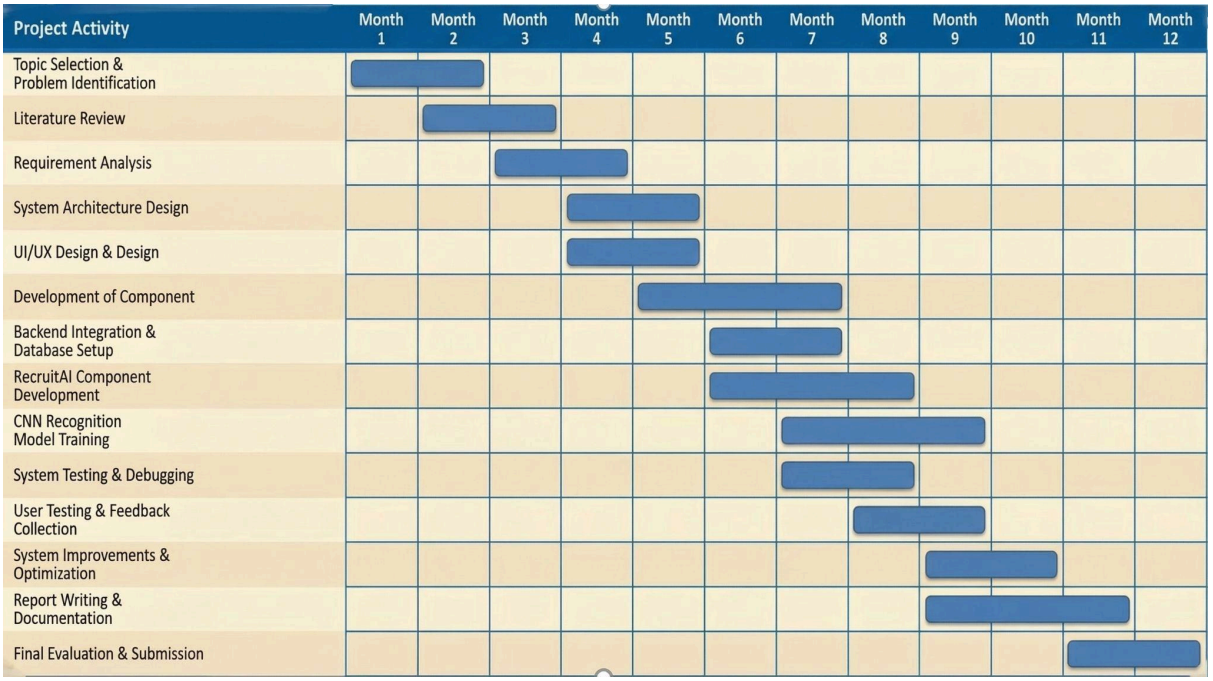
For Employers:

- Workforce planning: Develop internal career path models and identify development needs
- Retention: Employees will see career growth within your organization
- Recruitment: Prospective employees will see career growth within your organization
- Benchmarking: Compare your career paths to what is typical within your industry

8. WORK BREAKDOWN STRUCTURE (WBS)



9. GANTT CHART



10.REFERENCES LIST

[1] S. Kim, H. Jeon, J. Park, and S. W. Kim, "CAPER: Enhancing Career Trajectory Prediction using Temporal Knowledge Graph and Ternary Relationship," presented at ACM KDD, 2025. [Online]. Available: arXiv:2408.15620

[2] European Network of Public Employment Services, "Helping PES anticipate emerging challenges and future skills needs: the role of Labour Market and Skills Intelligence," European Commission, 2025. [Online]. Available: https://employment-social-affairs.ec.europa.eu

[3] X. Li, J. Wang, and Y. Zhang, "An approach to constructing a graph data repository for course recommendation based on IT career goals in the context of big data," in *Proc. IEEE International Conference on Big Data*, Osaka, Japan, 2022, pp. 1-8. [Online]. Available: IEEE Xplore.

[4] K. Vafa, "Modeling Worker Career Trajectories with Neural Sequence Models," presented at NeurIPS, 2021. [Online]. Available: https://neurips.cc/virtual/2021/38044

[5] AMiner, "CAPER: Enhancing Career Trajectory Prediction Using Temporal Knowledge Graph and Ternary Relationship," 2025. [Online]. Available: https://www.aminer.cn

[6] X. Xu, "Development of a Socio-Technical Deep Neural Network Framework for Personalized Career Guidance and Entrepreneurship Training: A Systems Approach," *International Journal of Information Technology and Systems Approach*, vol. 19, no. 1, pp. 1-19, 2025. [Online]. Available: IGI Global.

[7] K. Vafa, E. Palikot, T. Du, A. Kanodia, S. Athey, and D. Blei, "CAREER: Economic Prediction of Labor Sequence Data Under Distribution Shift," presented at NeurIPS Workshop on Distribution Shifts, 2022. [Online]. Available: https://nips.cc/virtual/2022/60530

[8] Z. Yanan, "Semantic-web–Enhanced Hybrid Learning for Career Planning: Ontology-driven Matching, Sequence Forecasting, and Closed-loop Optimization," *Journal of ICT Standardization*, vol. 13, no. 3, pp. 301–326, Dec. 2025. [Online]. Available: River Publishers.

[9] S. Gollapalli, X. Li, and S. C. Hui, "Generating Unified Candidate Skill Graph for Career Path Recommendation," in *Proc. IEEE International Conference on Data Mining*, Singapore, 2018, pp. 1-6. [Online]. Available: IEEE Xplore.

[10] M. Patton and W. McMahon, *Career Development and Systems Theory: Connecting Theory and Practice*, 3rd ed. Springer, 2014.